



MOIZ DAWOODI
O/A LEVEL PHYSICS

O' LEVEL PHYSICS

Momentum

1.6

STUDENT NAME



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PHYSICS WITH
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SYLLABUS OUTLINE:

1.6 Momentum

- 1 Define momentum as mass $\times$ velocity; recall and use the equation
 $p = mv$
- 2 Define impulse as force $\times$ time for which force acts; recall and use the equation
impulse $= F\Delta t = \Delta(mv)$
- 3 Apply the principle of the conservation of momentum to solve simple problems in one dimension
- 4 Define resultant force as the change in momentum per unit time; recall and use the equation

$$\text{resultant force} = \frac{\text{change in momentum}}{\text{time taken}}$$

$$F = \frac{\Delta p}{\Delta t}$$

MOMENTUM.

"The product of mass & velocity is called momentum."

FORMULA:

Momentum = Mass $\times$ Velocity

$$P = m v$$

Its a vector quantity. Its S.I unit is Ns .

IMPULSE:

When a steady force acts on mass ' m ' & its velocity changes from ' u ' to ' v ' in ' Δt ' time, then resultant force is given by:

$$\Sigma F = ma = m \frac{\Delta v}{\Delta t}$$

Hence,

$$F \Delta t = m \Delta v \quad \text{Where 'F}\Delta t\text{' is called an impulse.}$$

"Impulse is the product of force & time taken."

RESULTANT FORCE:

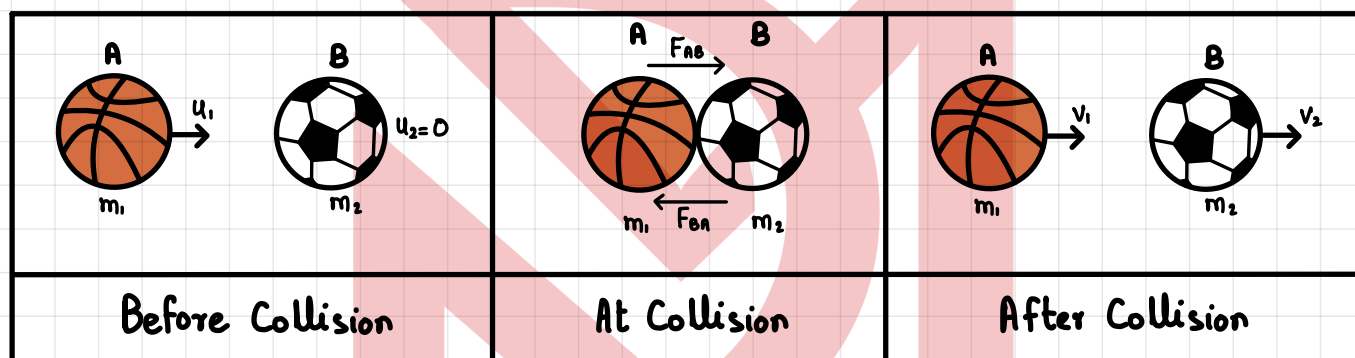
"Resultant force is the rate of change of momentum."

FORMULA: Resultant Force = $\frac{\text{Change in Momentum}}{\text{Time}}$

$$\Sigma F = \frac{\Delta P}{t}$$

PRINCIPLE OF CONSERVATION OF MOMENTUM.

"The total momentum before & after collision remains same under isolated conditions."



Consider two ball A & B having masses m_1 & m_2 respectively.

Ball A is moving with velocity ' u_1 ' towards ball B at rest. After sometime it collides with ball B. During collision, both ball applies force on each other according to Newton's 3RD Law. Ball A moves with velocity ' v_1 ' while ball B moves with velocity ' v_2 ' after collision.

Hence, expression can be found out by:

$$F_{AB} = -F_{BA} \Rightarrow \frac{\Delta P_{AB}}{t} = -\frac{\Delta P_{BA}}{t} \Rightarrow m_1 \Delta v_1 = -m_2 \Delta v_2$$

$$m_1 (v_1 - u_1) = -m_2 (v_2 - u_2) \Rightarrow m_1 v_1 - m_1 u_1 = -m_2 v_2 + m_2 u_2$$

$$m_1 v_1 + m_2 v_2 = m_1 u_1 + m_2 u_2$$

Total momentum after Collision = Total momentum before Collision

• In practical world, some energy loses as heat & sound during collision.



MATCH THE FOLLOWING



Momentum is the product

Total momentum,

Impulse is defined as:

Resultant force is the

before & after collision remains conserved under isolated conditions.

of mass & velocity.

rate of change of momentum

product of force & time taken.

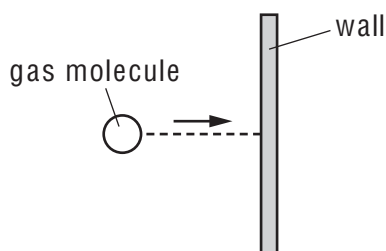
- 1 An object of mass 50 kg accelerates from a velocity of 2.0 m/s to a velocity of 10 m/s in the same direction.

What is the impulse provided to cause this acceleration?

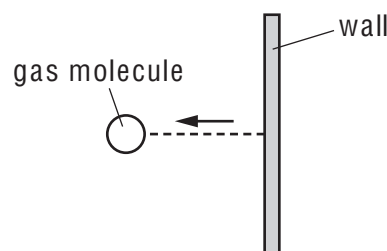
$$\text{Impulse} = m \Delta v = 50(10 - 2) = 50 \times 8$$

- A 250 N s B 400 N s C 850 N s D 2500 N s

- 2 A gas molecule strikes the wall of a container. The molecule rebounds with the same speed.



before hitting the wall



after hitting the wall

What happens to the kinetic energy and what happens to the momentum of the molecule?

	kinetic energy	momentum
A	changes	changes ✓
B	changes	stays the same
C	stays the same ✓	changes ✓
D	stays the same ✓	stays the same

- 3 Which quantity is measured in newton seconds (Ns)?

A impulse

P or Impulse

B moment

C power

D work done

- 4 During a paintball fight, a paint pellet of mass 150 g hits a stationary target with a speed of 220 m/s . It takes 0.025 s from the moment the pellet comes into contact with the wall until it flattens onto the wall.

$v = 0$

What is the force exerted as a result of the paintball "splat"?

A $1.3 \times 10^3 \text{ N}$

B 33 N

C $1.3 \times 10^6 \text{ N}$

D $33 \times 10^3 \text{ N}$

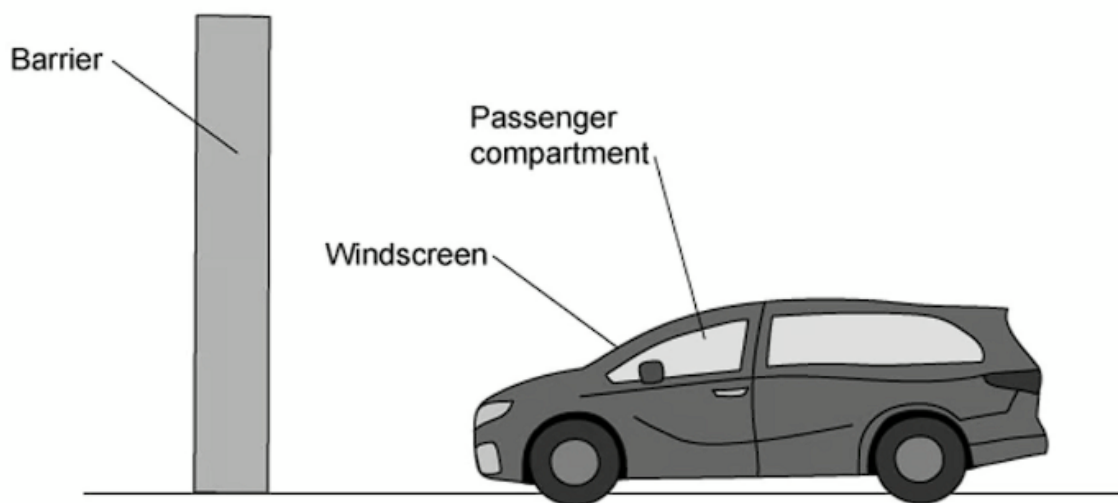
$$\Sigma F = \frac{\Delta P}{\Delta t} = \frac{m \Delta v}{\Delta t} = \frac{0.15 \times (220 - 0)}{0.025}$$

- 5 An object of mass 150 kg accelerates from a velocity of 5 m/s to a velocity of 10 m/s in the same direction.

What is the impulse provided to cause this acceleration?

$$\text{Impulse} = m \Delta v = 150 (10 - 5) = 150 \times 5$$

- A 750 Ns
B 1500 Ns
C 2250 Ns
D 7500 Ns
- 6 A passenger of mass 90 kg is involved in a minor car crash.



The car approaches a solid barrier at u 32 m/s. It crashes into the barrier and stops in 0.2s. $v = 0$
 Δt

Determine the impulse that must be applied to the car to bring it to rest.

- A 2.8 Ns
B 14 Ns
C 580 Ns
D 2900 Ns
- $$\text{Impulse} = m \Delta v = 90 \times (32 - 0) = 2880 \text{ Ns}$$

ANSWER KEY	
1	B
2	C
3	A
4	A
5	A
6	D

- 1 Fig. 1.1 shows a dummy of mass 70 kg used in a crash test to investigate the safety of a new car.

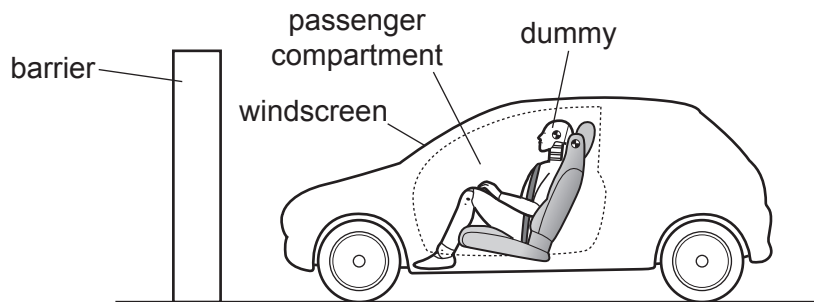


Fig. 1.1

The car approaches a solid barrier at 20 m/s. It crashes into the barrier and stops suddenly.

- (a) (i) Calculate the momentum of the dummy immediately before the crash.

momentum = [2]

- (ii) Determine the impulse that must be applied to the dummy to bring it to rest.

impulse = [1]

- (b)** In the crash test, the passenger compartment comes to rest in 0.20 s.

Calculate the deceleration of the passenger compartment.

deceleration = [2]

- (c)** The seat belt and air bag bring the dummy to rest so that it does not hit the windscreen.
The dummy has an average deceleration of 80 m/s^2 .

Calculate the average resultant force applied to the dummy, of mass 70 kg.

force = [2]

- (d)** The deceleration of the dummy is less than the deceleration of the passenger compartment.

Explain why this is of benefit for the safety of a passenger.

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..... [2]

- 2 Fig. 2.1 shows a hammer being used to drive a nail into a piece of wood.

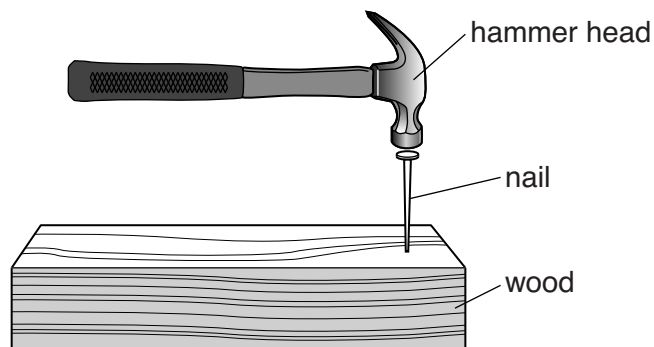


Fig. 2.1

The mass of the hammer head is 0.15 kg.

The speed of the hammer head when it hits the nail is 8.0 m/s.

The time for which the hammer head is in contact with the nail is 0.0015 s.

The hammer head stops after hitting the nail.

- (a) Calculate the change in momentum of the hammer head.

change in momentum =[2]

- (b) State the impulse given to the nail.

impulse =[1]

- (c) Calculate the average force between the hammer and the nail.

average force =[2]

- 3 The engine of an unpowered toy train is rolling at a constant speed on a level track, as shown in Fig. 3.1. The engine collides with a stationary toy truck, and joins with it.

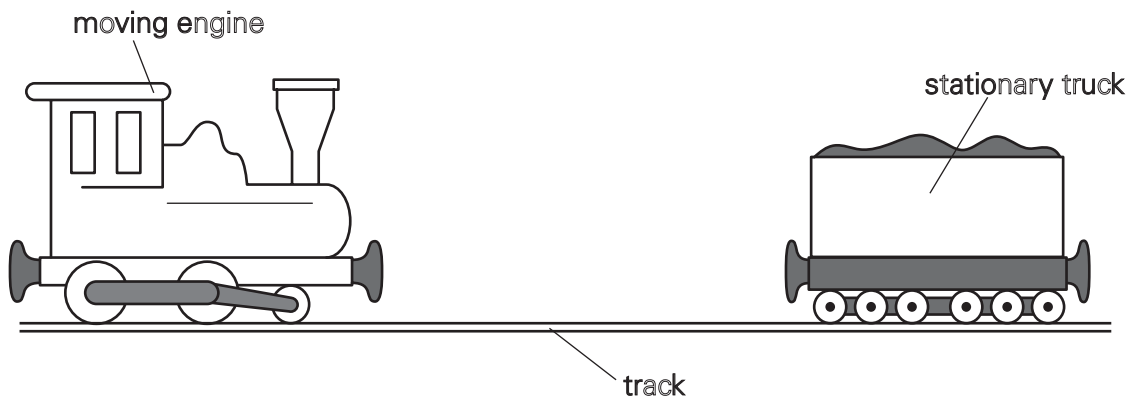


Fig. 3.1

Before the collision, the toy engine is travelling at 0.32 m/s . The mass of the engine is 0.50 kg .

- (a) Calculate the momentum of the toy engine before the collision.

momentum = [2]

- (b) The mass of the truck is 0.30 kg .

Using the principle of conservation of momentum, calculate the speed of the joined engine and truck immediately after the collision.

speed = [3]

Page 2	Mark Scheme	Syllabus	Paper

Question	Answer	Mark
1(a)(i)	(momentum =) mv OR 70×20 $= 1400 \text{ kg m/s}$ OR Ns	C1 A1
a)(ii)	same numerical answer as (a)(i) with either unit OR 1400 kg m/s	B1
(b)	(a =) change of velocity/time OR $(v - u)/t$ OR $20/0.2$ 100 m/s^2	C1 A1
(c)	($F =$) ma OR 70×80 5600 N	C1 A1
(d)	Force/impact on passenger or dummy less (than without seat belt/airbag) Passenger less likely to be injured/hurt/damaged	M1 A1

Page 2	Mark Scheme	Syllabus	Paper

- 2 (a) $mv - mu$ OR $m(v - u)$ OR mv OR 0.15×8.0 C1
 1.2 Ns or kg m/s A
 (b) 1.2 Ns or kg m/s B
 (c) $F = (mv - mu)/t$ OR $F = mv/t$ OR impulse/ t OR $1.2/0.0015$
 800 N A1
 OR
 $(F =) ma$ OR $m[(v - u)/t]$ OR $0.15 \times 8/0.0015$
 800 N (A1)

Page 2	Mark Scheme	Syllabus	Paper

- 3 (a) $p = mv$ in any form, words or symbols [1]
 0.16 kg m/s OR Ns [1]
 (b) use of principle of conservation of momentum in words, symbols or numbers [1]
 use of combined mass $0.5(0) + 0.3(0)$ OR $0.8(0)$ (kg) [1]
 $0.2(0) \text{ m/s}$ [1]